RF-Based Drone Identification Using Machine Learning: Presence, Family and Operating-Condition Classification

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***Abstract*—Drones communicate with their controllers over radio-frequency (RF) links. Listening to those signals is a passive way to detect and identify a drone that works without line of sight or good lighting, unlike cameras and radar. This project builds a machine-learning pipeline on the public DroneRF dataset that answers three questions about a recorded RF segment. Is a drone present. Which drone family is it. Which recorded operating condition is most likely. Raw captures from the 2.4 GHz ISM band are cut into segments and turned into a 2048-point power-spectrum feature vector using the Fast Fourier Transform over the low and high RF sub-bands. We compare four classical classifiers on each task. They are Logistic Regression, Random Forest, Support Vector Machine with an RBF kernel, and XGBoost. The evaluation groups every segment of one recording into the same fold so the model never sees its test recordings during training. Drone presence reaches a macro-F1 of 1.000. Drone family (background, Parrot Bebop, Parrot AR, DJI Phantom) reaches 1.000 for XGBoost. The harder 10-class operating-condition task reaches 0.492 for XGBoost under 5-fold cross-validation, well above the roughly 0.37 of the linear and bagging models. A noise test shows family accuracy drops from 0.900 on clean data to 0.294 at -5 dB SNR, which measures how much the shared Wi-Fi and Bluetooth traffic in the band hurts the models. The system is an interpretable, low-cost tool for studying the dataset, not a deployed counter-drone system.**

***Index Terms*—Drone detection, RF fingerprinting, UAV identification, machine learning, DroneRF, XGBoost, feature extraction, power spectral density, data leakage.**

I. Introduction

Drones are now cheap and everywhere, and their misuse creates privacy, safety and security problems around airports, public events and restricted areas. Detecting and identifying an intruding drone reliably is the problem this project works on.

Cameras and microphones need line of sight and good conditions, and active radar can be expensive. Drones solve part of the problem for us. They keep talking to their flight controllers over wireless RF links, almost always in the 2.4 GHz industrial, scientific and medical (ISM) band. Listening to those signals detects a drone passively, without its cooperation and without line of sight.

Knowing only that a drone is present does not help much. A useful system should also say which drone it is and, if possible, what it is doing, be-

cause different models and flight modes carry different risks. This project builds a machine-learning pipeline that uses recorded DroneRF signals for three classification tasks of rising difficulty, presence, family and operating condition, and compares four classical machine-learning algorithms on each one.

II. Overview

The system reads recorded RF segments and produces three predictions. The three targets are defined as

$$\text{presence} \in \{0, 1\}, \quad \text{family} \in \{0, 1, 2, 3\},$$
$$\text{mode} \in \{0, 1, \ldots, 9\}.$$

Presence is drone against no-drone. Family separates background from three drone types. Mode names the specific recorded operating condition.

The pipeline runs in order as
Raw RF captures → segmentation → FFT feature extraction → label decoding → training of four classifiers → leakage-aware evaluation.

The data comes from one public source, the DroneRF dataset [1], [2]. A 5-digit BUI code in each filename carries all three labels, so no manual labelling is needed. Every target has a small number of discrete classes, so each task is a classification problem, not a regression problem.

III. Problem Statement

One sensor cannot read drone presence reliably in all conditions, yet security and airspace management depend on knowing it. RF sensing is passive and works without line of sight, but the 2.4 GHz band is shared with Wi-Fi and Bluetooth, so the recorded signals are noisy.

A second problem is easy to miss. Many segments cut from one physical recording are almost identical. If those segments land in both the training set and the test set, the model effectively studies the test data before the exam, and the reported accuracy is too good to believe. A fair system needs an evaluation that keeps all segments of a recording on the same side of the split.

The proposed system classifies DroneRF segments with classical machine learning and tests the trained models under a recording-grouped split and under added noise.

IV. Literature Survey

RF-based drone detection and identification with machine learning has a large literature. Early work built the dataset and a deep-learning baseline. Later work studied feature engineering, interference robustness and lightweight deployment.

A. Dataset and Deep-Learning Baseline

Al-Sa'd et al. [1] released the DroneRF dataset and trained deep neural networks for detection, type and mode classification. That paper fixed the three-task structure this project follows. The data descriptor [2] documents the file layout and BUI labelling.

B. Classical Machine Learning

Al-Emadi et al. [3] applied XGBoost to the same three tasks and reported average accuracies of 99.96%, 90.73% and 70.09% for detection, type and mode. The mode accuracy in particular shows how hard that task is, which matches our result. Ezuma et al. [4] classified UAV controllers from RF fingerprints using statistical features with classifiers such as k-nearest neighbours.

C. Interference and Robustness

Ezuma et al. [5] studied detection and classification under Wi-Fi and Bluetooth interference. Recent journal work keeps this focus, covering the robustness of RF UAV detectors [7], ensemble fingerprinting under jamming [8], and low-SNR recognition with complex-valued networks [9].

D. Feature Extraction

Medaiyese et al. [6] used wavelet-transform features. The LUMEN framework [10] used power-spectral-density features close to the ones used here. Both support the spectral feature choice of this project.

E. Comparative Analysis

Table I places representative work next to the proposed system.

TABLE I
LITERATURE COMPARISON

Reference	Method	Year	Focus
Al-Sa'd et al. [1]	Deep neural network	2019	DroneRF dataset, 3 tasks
Al-Emadi et al. [3]	XGBoost	2020	Detect, type, mode
Ezuma et al. [5]	ML with interference	2020	Wi-Fi, BT robustness
Medaiyese et al. [6]	Wavelet and ML	2022	Feature extraction
LUMEN [10]	PSD, lightweight net	2026	Edge deployment
Proposed	**4 classical ML, FFT features**	**2026**	**Leakage-aware 3-task comparison**

F. Research Gap

Most existing work either uses deep learning or reports high accuracy without checking for segment-level data leakage. This project fills that gap. It compares four interpretable classical models across all three tasks under a recording-grouped split, and it measures noise robustness directly.

V. Libraries and Tools Used

Python 3.11 is the core language for preprocessing, feature extraction, training, evaluation and plotting.

- **NumPy** runs the FFT, array operations and the power-spectrum feature computation.
- **Pandas** loads and organizes the feature table and writes it to a columnar Parquet file.
- **scikit-learn** provides Logistic Regression, Random Forest, SVM, the grouped split (`GroupKFold` and `GroupShuffleSplit`) and the metrics (precision, recall, F1, confusion matrix).
- **XGBoost** provides the gradient-boosted tree classifier.
- **Matplotlib** draws the confusion-matrix heatmaps and the noise-degradation curve.

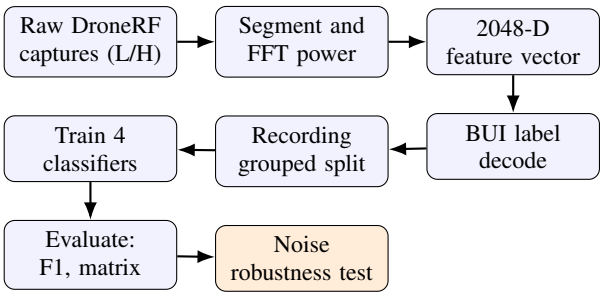


Fig. 1. System flow diagram.

TABLE II
BUI CODE TO LABELS

BUI code(s)	Family	Meaning
00000	0	Background, no drone
10000 to 10011	1	Bebop, 4 modes
10100 to 10111	2	AR drone, 4 modes
11000	3	Phantom, 1 mode

- **Kaggle API** downloads the DroneRF dataset reproducibly.

VI. System Design and Methodology

Fig. 1 shows the pipeline.

A. Raw Data

The DroneRF dataset holds recorded RF activity for background (no drone) and three drones, the Parrot Bebop, Parrot AR and DJI Phantom. Each capture is two CSV files, a low sub-band (`L`) and a high sub-band (`H`) of the 2.4 GHz band. There are 227 recording pairs.

B. Label Decoding

Each filename starts with a 5-digit BUI code that encodes all three targets at once, shown in Table II. Presence is 0 for background and 1 otherwise. Family maps the code to one of four classes. Mode is a separate index for each of the ten codes.

C. Feature Extraction

Each capture is split into non-overlapping segments of $L = 100{,}000$ samples. For a segment x, we remove the mean and take a 2048-point FFT magnitude, keeping the upper half ($M/2 = 1024$ bins),

$$X_f = \left| \mathrm{FFT}_M(x - \bar{x}) \right|, \qquad f = \tfrac{M}{2} + 1, \ldots, M.$$

TABLE III
MACRO-F1 BY MODEL AND TASK

Task	LR	RF	SVM	XGBoost
Presence	1.000	0.978	1.000	1.000
Family	0.999	0.898	0.977	**1.000**
Mode (5-fold)	0.375	0.378	0.160*	**0.492**

*SVM trained on a 5000-sample subset per fold.

The low- and high-band spectra are joined with a continuity scaling and squared to give a power spectrum. This produces a 2048-dimensional feature vector per segment. The full dataset gives 22,700 feature vectors.

D. Leakage-Aware Split

All segments of one capture are almost identical, so the evaluation keeps them together using `GroupKFold` with 5 folds, or a single 75/25 grouped split. No recording appears in both the training and test sets.

E. Classifiers

Four models are trained and compared on each task. Logistic Regression is the linear baseline. Random Forest is a bagging ensemble of trees. SVM with an RBF kernel handles curved class boundaries and trains on a 5000-sample subset to stay fast. XGBoost is a gradient-boosted tree ensemble.

F. Evaluation Metrics

For each model and task we report macro-averaged precision, recall and F1, with a confusion matrix. Macro-F1 is the headline number because the classes are imbalanced,

$$\text{F1} = \frac{2\,PR}{P+R}, \qquad \text{macro-F1} = \frac{1}{K}\sum_{k=1}^{K} \text{F1}_k.$$

VII. RESULTS AND DISCUSSION

Table III reports macro-F1 for every model and task. Presence and family use a single grouped split. The harder mode task is reported as mean and standard deviation over 5 folds.

Every model solves presence detection. XGBoost also solves family classification, and its confusion matrix (Fig. 2) is a clean diagonal. The 10-class mode task is much harder. XGBoost reaches 0.492, well above the linear and bagging baselines near 0.37, so it is the best model on the task that actually separates the methods.

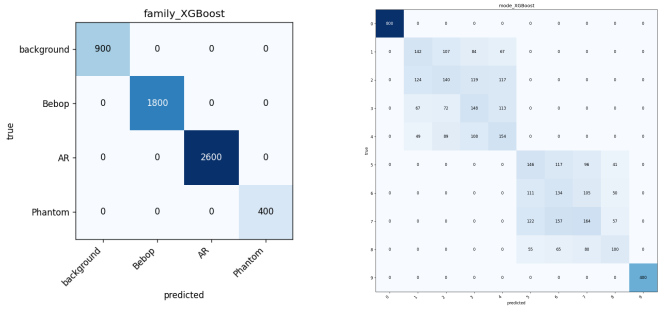


Fig. 2. XGBoost confusion matrices. Family on the left is a clean diagonal. Mode on the right is the 10-class task.

TABLE IV
FAMILY MACRO-F1 VS. SNR (RANDOM FOREST)

SNR	clean	20 dB	10 dB	5 dB	0 dB	−5 dB
Macro-F1	0.900	0.496	0.424	0.376	0.320	0.294

A. Noise Robustness

To test the interference concern of the 2.4 GHz band, we added Gaussian noise to the test features at falling SNR for the family task (Table IV, Fig. 3). Accuracy falls from 0.900 on clean data to about 0.50 at 20 dB and 0.29 at -5 dB. Spectral features alone are not robust to heavy noise, so a deployed system would need explicit interference handling.

B. Fair-Evaluation Note

The near-perfect presence and family scores are honest under the recording-grouped split, though some correlation between captures of the same mode may still remain. The small 5-fold standard deviation of ± 0.009 says the results are stable, not the product of one lucky split.

VIII. CONCLUSION

This project built and evaluated a classical machine-learning pipeline for RF-based drone identification on the DroneRF dataset. Four models were compared across presence, family and operating-condition classification under a leakage-aware split. Presence and family classify reliably. The 10-class operating-condition task stays hard, with XGBoost

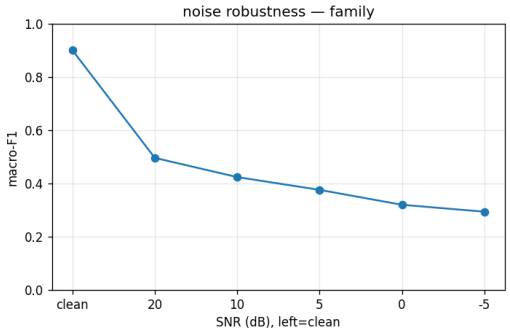


Fig. 3. Noise-robustness curve for the family task.

giving the best macro-F1 of 0.492. A noise test measured how fast the spectral features degrade under interference. Interpretable, low-cost classical models stay competitive on RF drone classification. The main directions for future work are the operating-condition task and noise robustness.

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